Lab: Neural Systems Simulation

> **Learning Goal:** Explore neural dynamics through hands-on analysis of oscillations and attractor behavior.

Part 1: Neural Oscillation Analysis

Exercise: Watch or read about EEG frequency bands (alpha, beta, gamma, theta, delta).

For each frequency band, fill in:

Band	Frequency	When is it prominent?	What cognitive function does it support?
Delta	0.5-4 Hz		
Theta	4-8 Hz		
Alpha	8-13 Hz		
Beta	13-30 Hz		
Gamma	30-100 Hz		

Write your response here

Part 2: Attractors in Everyday Life

> **Learning Goal:** Develop intuition for attractor dynamics through non-neural examples.

Exercise: For each system, identify the attractor(s) and classify them (fixed-point, limit-cycle, or complex):

1. A pendulum swinging in air (friction present)
2. A beating heart
3. A marble in a bowl

Now apply this to neural systems: 4. Resting-state brain activity 5. A seizure 6. The pattern of neural activity when you recall your phone number

Write your response here

Part 3: Excitation-Inhibition Balance

> **Learning Goal:** Understand how populations produce oscillations through E-I balance.

Thought experiment: Imagine a small network of 10 excitatory and 10 inhibitory neurons, all connected to each other.

1. If only excitatory neurons were present, what would happen? (Hint: think runaway excitation)
2. If only inhibitory neurons were present, what would happen?
3. With both, how does the interplay produce oscillations?
4. What happens when the balance is disrupted? (Connect to epilepsy)

Write your response here

Part 4: The Brain's Markov Blanket at Multiple Scales

> **Learning Goal:** Apply the Markov Blanket concept to different scales of neural organization.

Exercise: For each level, identify: boundary, sensory inputs, active outputs, internal dynamics.

Scale	Example	Boundary	Sensory Inputs	Active Outputs	Internal Dynamics
Single neuron	Pyramidal cell				
Local circuit	Cortical column				
Brain region	Visual cortex (V1)				
Whole brain					

Write your response here

Part 5: Reflection

In 150 words, reflect: Why is it useful to think of the brain as a dynamical system rather than a computer? What does the dynamical systems perspective capture that the "brain as computer" metaphor misses?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Data interpretation	Neural oscillation frequency bands
2	Attractor classification	Attractors in neural and non-neural systems
3	Mechanistic reasoning	Excitation-inhibition balance
4	Multi-scale analysis	Markov Blankets at neural scales
5	Critical reflection	Dynamical systems vs. computational metaphors